Arten

Edge Object Detection, Counting & Attributes

Offline edge detection and counting for safer streets

Namma AI Hackathon 2026 · Problem Statement 2 — AI for Safer Communities

Team: [Team name] · [Member 1], [Member 2], [Member 3] · [College / organisation]

One line

Arten turns a 15-second phone clip of any street into an offline, privacy-preserving count of pedestrians and vehicles on a Raspberry Pi — with on-device direction, dwell and coarse colour attributes. No cloud, no GPU and no identities.

1. The problem

Road traffic injuries are the leading cause of death for children and young adults aged 5–29 worldwide (WHO). India reported **1,68,491 road deaths in 2022** (MoRTH); pedestrians were **19.5%** of the people killed and two-wheeler riders **44.5%**. These are not abstract numbers. They are the school gate, the market junction and the bus stop.

The data needed to make a specific street safer is simple: *how many people cross here, when, and alongside how much traffic?* Yet that data is usually locked behind expensive cameras, proprietary dashboards or continuous cloud connectivity. A ward officer, school principal or resident welfare association cannot justify a surveillance platform. They can justify a sub-₹10,000 computer and a 15-second clip.

2. The insight

Count movement, not people.

We do not need face recognition, licence plates or persistent identities to make a road safer. Aggregate counts at different times of day are enough to prioritise a crossing guard, a zebra crossing, a speed hump or a footpath repair. That makes the problem a natural fit for a small, offline edge detector — and it lets us design for privacy from the first line of code.

3. What we will build

A functional PoC that satisfies every mandatory deliverable:

1. **Input:** one organiser-provided 10–20 second clip, or 10–20 images. The committed sample clip is 15 seconds / 150 frames.
2. **Detection:** six COCO categories — person, bicycle, car, motorcycle, bus, truck — with per-frame counts for each.
3. **Evidence:** terminal table, `counts.csv`, `detections.jsonl`, annotated frames, a contact sheet and an optional annotated video.

4. **Performance:** model-only and full-pipeline latency (mean, p50, p95), throughput, model size, checksum and process memory.
5. **Explanation:** a two-minute societal-impact pitch and a technical report with limitations.

Optional refinements that make the counts more useful without overbuilding:

- A lightweight IoU/velocity tracker for **indicative unique counts** across a clip. Clearly labelled as indicative, never presented as production MOT.
- **Observed person-seconds / vehicle-seconds** for the clip, so two clips of different lengths can be compared without pretending to extrapolate a full day.
- **On-device attributes** alongside each tracked detection: image-plane direction, dwell time, relative speed and optional coarse vehicle colour. Derived from the same tracker and classical CV, not a second model; no identity attributes.
- A **self-contained run report** (`REPORT.md`) generated from the same JSON that drives the terminal output, so every number is traceable.

4. Why ExecuTorch on a Raspberry Pi

The primary artifact is **Arm's pre-optimized YOLO11n INT8 ExecuTorch model** (4.73 MB), not a PyTorch script running on a laptop.

Property	Value
Model	YOLO11n, 80-class COCO detector
Quantization	INT8 static PTQ; stem conv and detection head in FP32
Runtime	ExecuTorch 1.1.0, XNNPACK + KleidiAI, CPU only
Pi 5 reference latency	**134.10 ms p50 · 7.46 FPS**
Pi 5 reference peak memory	**59.17 MB**
COCO accuracy	**38.90% mAP@0.5:0.95 · 54.66% mAP@0.5**
Model size	**4.73 MB**

At 7.46 FPS, a 15-second / 150-frame clip takes about 20 seconds of model time on a Pi 5. That is comfortably inside a one-day hackathon, and fast enough to process a handful of clips during a school-zone audit. There is no GPU, no accelerator and no network dependency — just the Pi's Cortex-A76 cores and XNNPACK.

We deliberately chose a **pre-optimized, verified artifact** over a larger model that would look better on paper but fail on the target. The model card publishes p50, p90, p99, load time, memory and accuracy, so the proposal's performance claims are anchored to the target device, not a laptop.

5. How it works

```
clip/images
   → letterbox 640×640 (RGB, /255, 114 pad)
   → ExecuTorch .pte (YOLO11n INT8, XNNPACK)
   → decode 4 xywh + 80 classes
   → confidence filter + per-class NMS
   → per-frame counts (+ optional IoU tracker)
   → CSV / JSONL / annotated frames / video / latency report
```

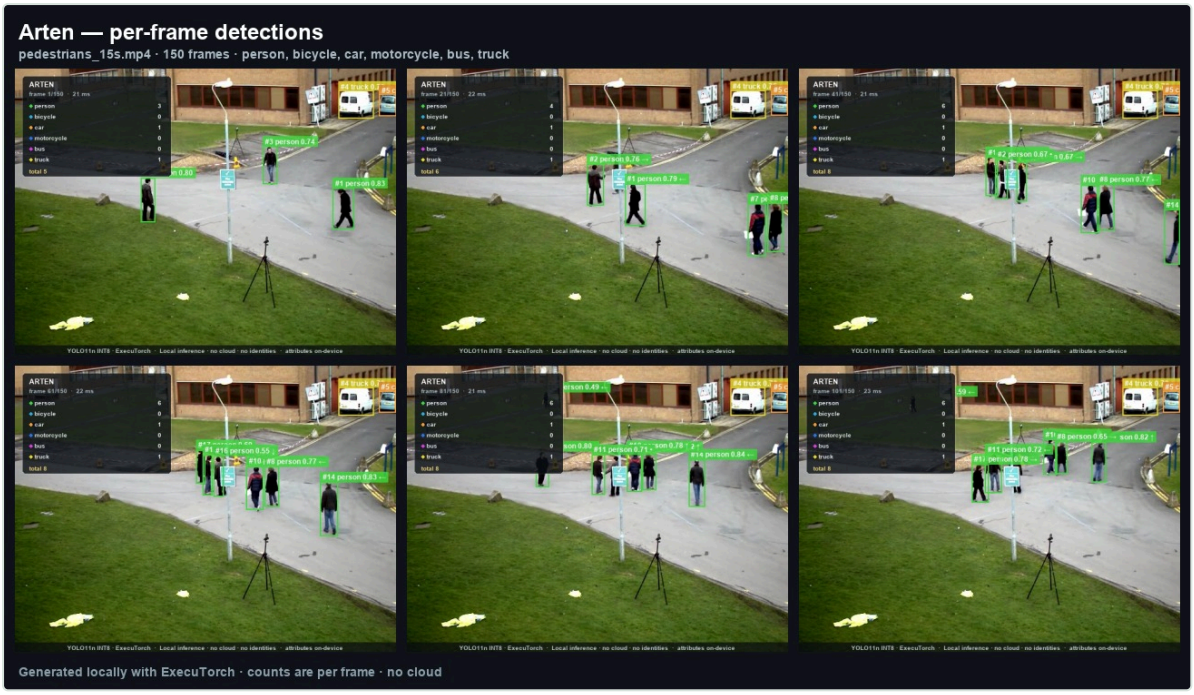
The pipeline is a small, tested Python package. Preprocessing is matched to the export contract (including the 114-grey pad and no mean/std normalisation); getting that wrong silently shifts boxes. Decode and NMS are NumPy, so the Pi install stays small. The CLI accepts any YOLO-style `.pte` with an `[N, 84]` or `[N, 85]` output, so if the organisers approve a different model, we change one argument — not the application.

6. What is already built

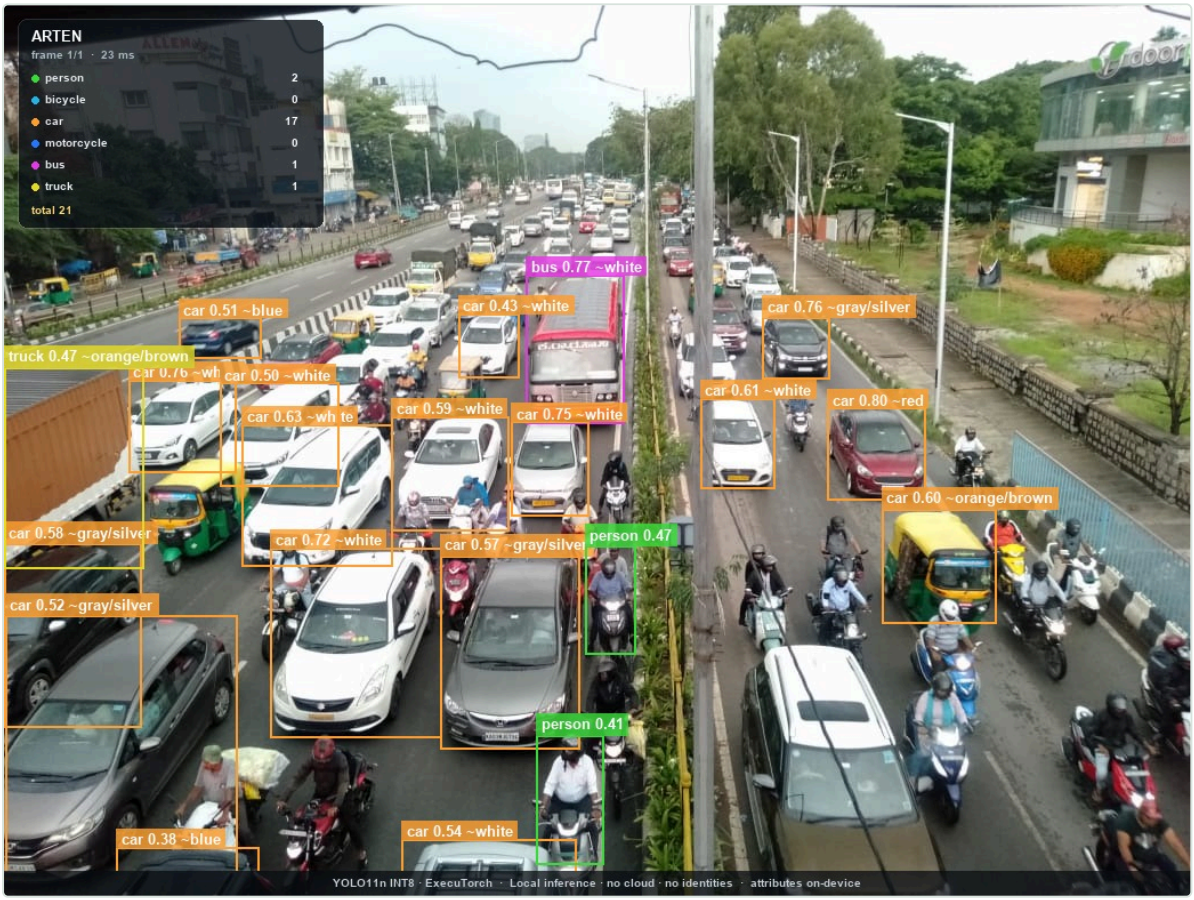
This is not a slideware proposal. The repository already contains:

- A working ExecuTorch pipeline tested on the committed clip and two Bengaluru images.
- **29 passing unit tests** for letterboxing, NMS, decode, tracker identity, attributes, zone metrics and latency statistics.
- A one-command Raspberry Pi setup script with a pinned runtime, checksum verification and a smoke test.
- A reproducible evidence bundle: per-frame CSV, detections JSONL, annotated frames, contact sheet, annotated video and a generated Markdown report.
- A dev-host benchmark: **20.34 ms mean model-only, 49.16 FPS** on Apple Silicon (explicitly labelled as not a Pi measurement).
- A smoke test of the **exact Pi runtime stack** (ExecuTorch 1.1.0 + torch 2.10.0) on the dev host: 3 frames end-to-end, same output shape and counts.
- Attribute evidence: on the committed clip, **18 confirmed tracks** split **9 left / 4 right / 2 stationary / 2 down / 1 up**, mean dwell **4.5 s**; on the dense traffic image, **17 cars** split into **9 white, 4 gray/silver, 2 blue, 1 red, 1 orange/brown** (coarse HSV estimate).
- A 21-second demo video with title card, live detections, counts, attribute labels and an end card.

At the finals we will run the same commands on the organiser clip on the Pi, replace the dev-host row with on-device numbers, and show the raw CSV next to the annotated frames. The one-day event becomes a deployment and validation exercise, not a from-scratch build.



Per-frame pedestrian detections



Dense Bengaluru traffic stress test

7. Practical application: the school-zone safety audit

Who benefits: a school management committee, a ward road-safety officer or a resident welfare association.

The workflow:

1. Record three 15-second clips at the school gate: morning drop-off, afternoon dismissal and a quiet hour.
2. Run each clip on the Pi. No internet, no footage leaves the school.
3. Compare peak concurrent pedestrians, vehicle counts and observed person-seconds across the three clips.
4. If dismissal shows a high pedestrian count with a high vehicle count, that is evidence for a crossing guard at 3:30 pm, a zebra crossing, or a no-parking zone within 20 metres of the gate.

The output is not a dashboard. It is a one-page count table and a handful of annotated frames that a non-technical committee can read in two minutes.

8. Six-hour execution plan at the finals

Time	Work	Output
0:00–0:45	Flash/verify Pi, run `setup_pi.sh`, `doctor`, load model	Green environment
0:45–1:30	Run organiser clip, verify counts and annotated frames	First real result
1:30–2:30	Tune confidence threshold on 2–3 frames, freeze settings	Defensible config
2:30–3:30	Run full clip, benchmark 50 runs, save evidence bundle	CSV/JSON/video
3:30–4:30	Fill report tables, replace dev-host numbers, build contact sheet	Final report
4:30–5:30	Rehearse 2-minute pitch, prepare backup USB with outputs	Pitch ready
5:30–6:00	Buffer: rerun, fix, export	Submission

Fallbacks: prebuilt INT8 model committed; FP32 Ultralytics export script if needed; all evidence pre-generated on the dev host on a USB stick; the pipeline runs on x86_64 too if the Pi fails.

9. Risks and mitigations

Risk	Mitigation
ExecuTorch/PyTorch install is slow or fails on Pi	Pin known-good versions in `requirements-pi.txt`; setup script is idempotent; pre-download wheels on the dev host as a backup
Model misses small/occluded objects	Report it honestly; use close-range clips for the demo; document 640 input and tiling as future work
Auto-rickshaws are not a COCO class	Call it out as a limitation; do not fake a category; propose fine-tuning on Indian roads as next step
Organiser clip is a different codec/resolution	OpenCV handles common codecs; `read_image` honours EXIF; `--stride` and `--max-frames` control cost
Counts are challenged	Show per-frame CSV, annotated frames and the exact command; never claim accuracy without ground truth
Pi 4 instead of Pi 5	The pipeline runs on aarch64 Pi 4; report lower throughput honestly; Pi 5 is the reference target

10. Responsible AI

- **No identity extraction.** No face recognition, no licence plates, no re-identification. The default output is aggregate counts.
- **Local by construction.** No network calls in the inference path. The demo can run with the Wi-Fi disabled.
- **Aggregate, not enforcement.** The tool is a decision-support aid for infrastructure planning, not a policing or enforcement system.
- **Honest accuracy.** We report COCO mAP from the model card, not a fabricated accuracy on the organiser clip. We show failure cases, including the dense Bengaluru scene where motorcycles are missed.
- **Licence compliance.** YOLO11n and the Arm `.pte` are AGPL-3.0; the repository is AGPL-3.0 and the model notice is included.

11. Team

Member	Role	Relevant experience
[Member 1]	Edge/runtime, Pi deployment	[fill]
[Member 2]	Computer vision, evaluation	[fill]
[Member 3]	Product, societal application, pitch	[fill]

Appendix — mandatory deliverable map

Requirement	Where
Process 10–20 s clip or 10–20 images	`data/samples/pedestrians_15s.mp4` ; `arten run`
Detect at least one category + per-frame counts	`counts.csv` , `summary.txt` , HUD
Category + counts + 2 sample outputs	`results/demo_pedestrians/contact_sheet.jpg` , `results/demo_traffic/annotated/bengaluru_traffic.jpg` , `results/demo_street/annotated/bengaluru_street.jpg`
Average inference time, model, throughput	`summary.json` , `summary.txt` , `benchmark_dev.json`
2-minute societal explanation	`docs/pitch-2min.md`

Links: GitHub https://github.com/sujalmh/arten-edge-ai · Demo video https://github.com/sujalmh/arten-edge-ai/releases/download/v1.0.0/demo.mp4 · Technical report `docs/report.md` · Research notes `docs/research.md`